



PHENIKAA UNIVERSITY
Faculty of Computer Science

Software Analysis and Design

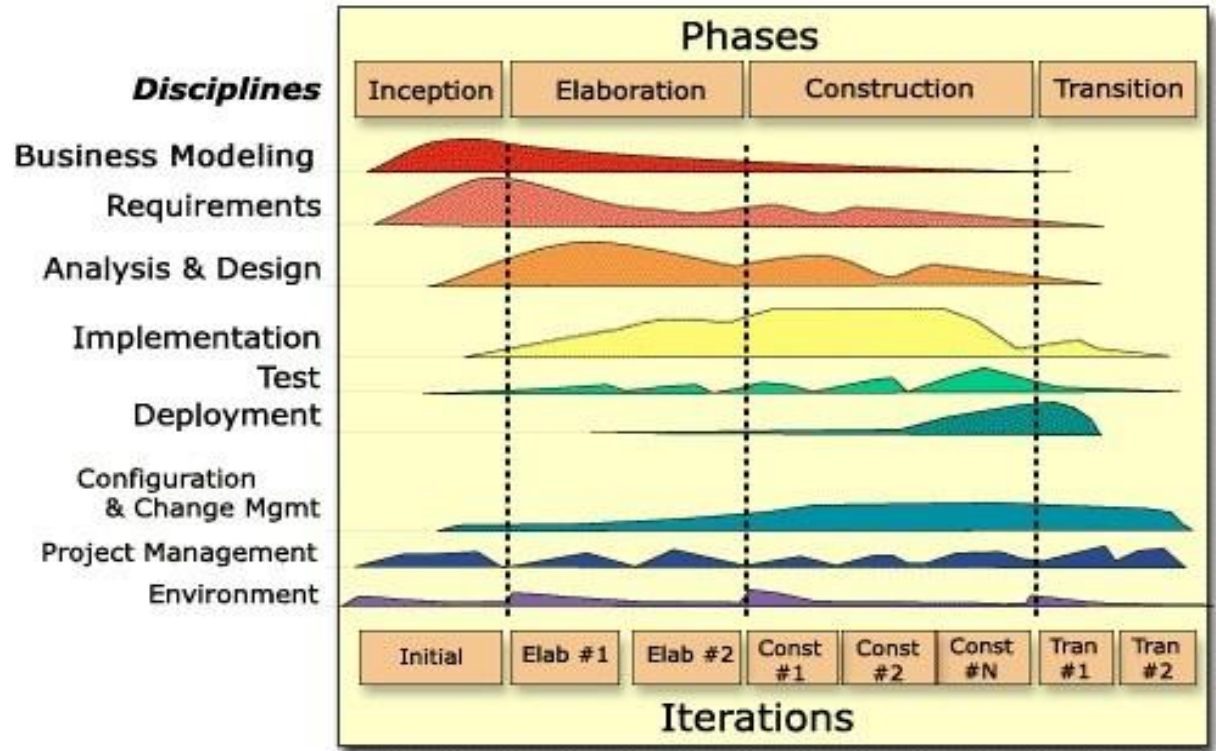
Module 10: Review

- ❖ Requirement Analysis
- ❖ Analysis and Design
- ❖ UML Diagrams

Purpose of Requirement Analysis



- ❖ Establish and maintain an agreement between customers and other stakeholders on what the system should do
 - ❖ Don't say how to do it
- ❖ Give system developers a better understanding of system requirements



Purpose of Requirement Analysis

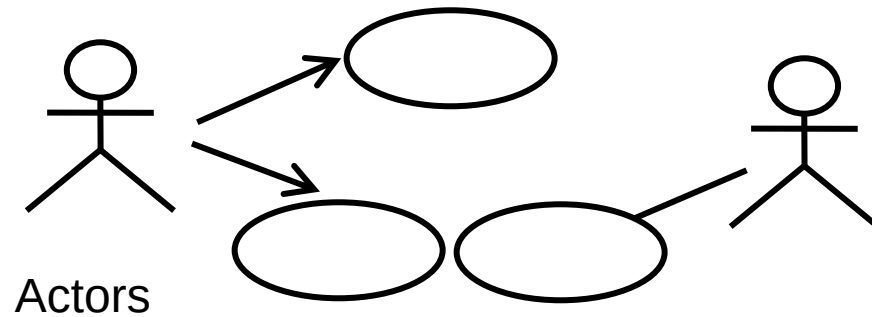


- ❖ Define what the system will and will NOT do
- ❖ Provide basic information for project development planning
- ❖ Provides the basis for estimating the cost and time to develop the system
- ❖ Identify the user interface of the system, capture the requirements and purposes of the user
- ❖ As input to the next phase of the project: Analysis and Design

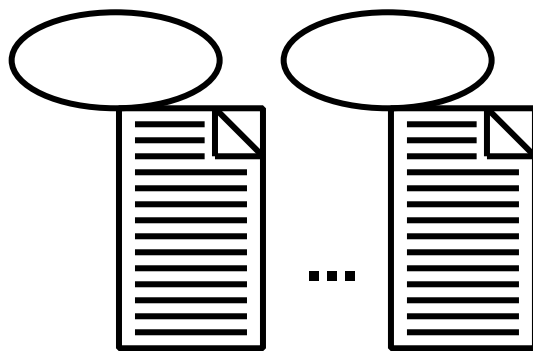
Relevant Requirements Artifacts



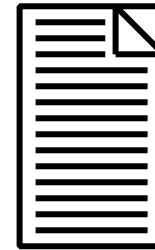
Use-Case Model



Use Cases



Use-Case Specifications



Glossary



Supplementary
Specification

- ❖ Use cases defined for a system are the basis for the entire development process.
- ❖ Benefits of use cases:
 - ❖ Concise, simple, and understandable by a wide range of stakeholders.
 - ❖ Help synchronize the content of different models

Content



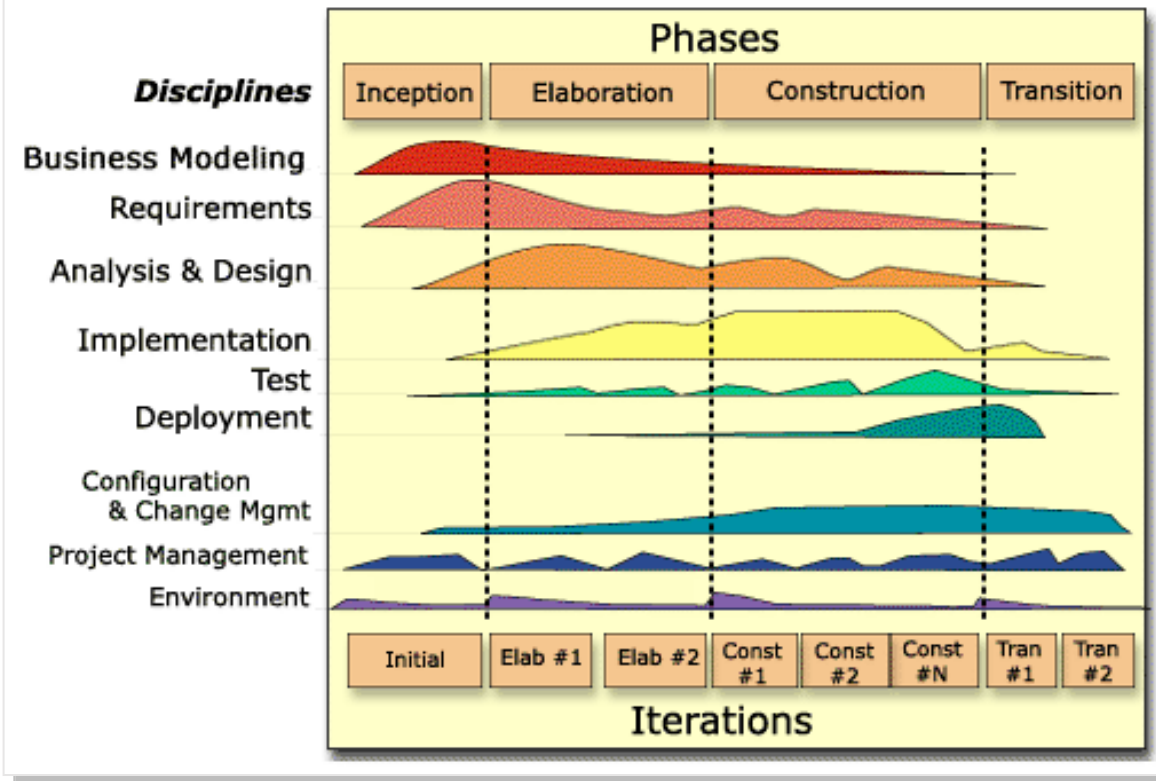
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Analysis and Design in Context

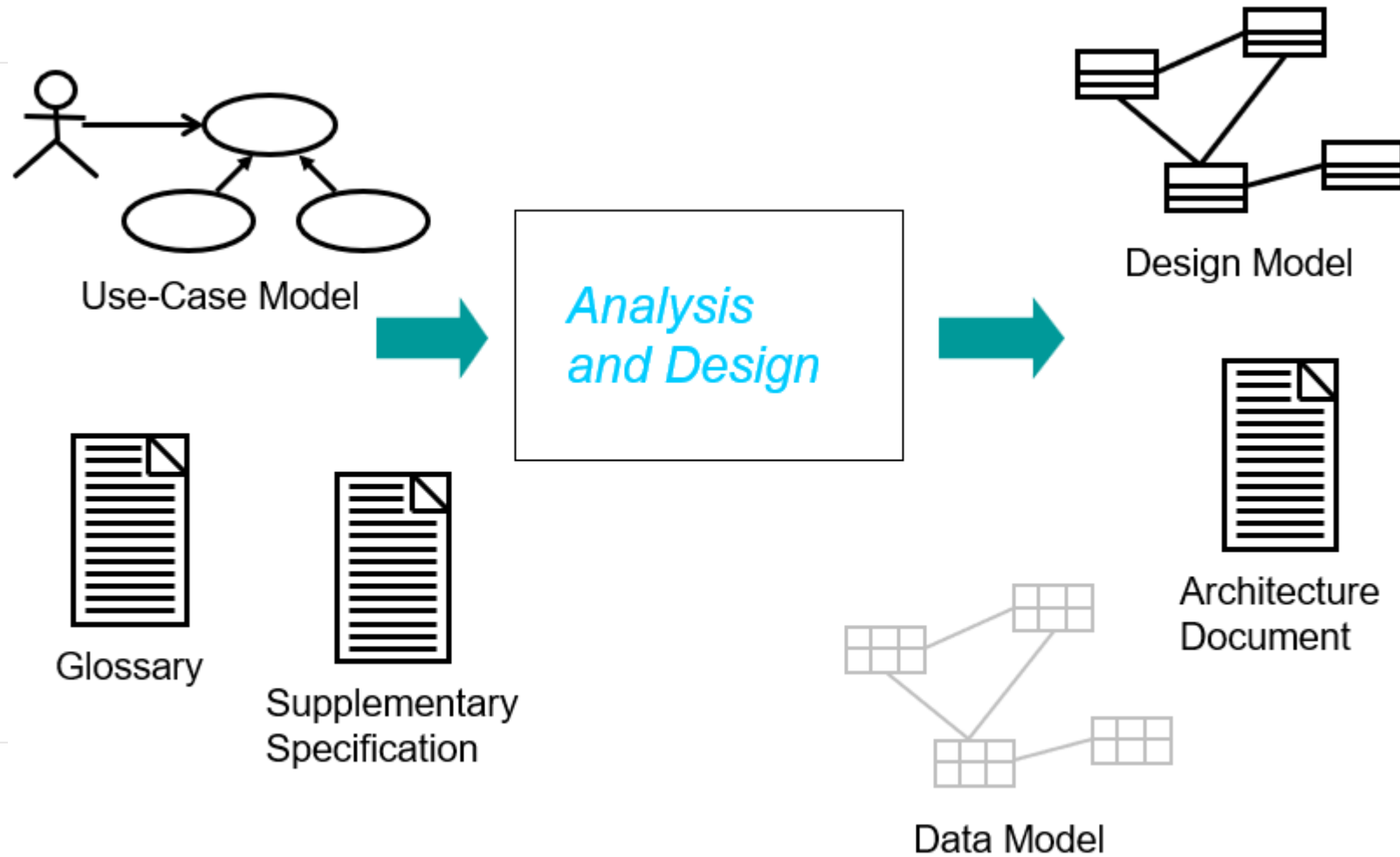


The purposes of Analysis and Design are to:

- Transform the requirements into a design of the system-to-be.
- Evolve a robust architecture for the system.
- Adapt the design to match the implementation environment, designing it for performance.



Analysis and Design Overview

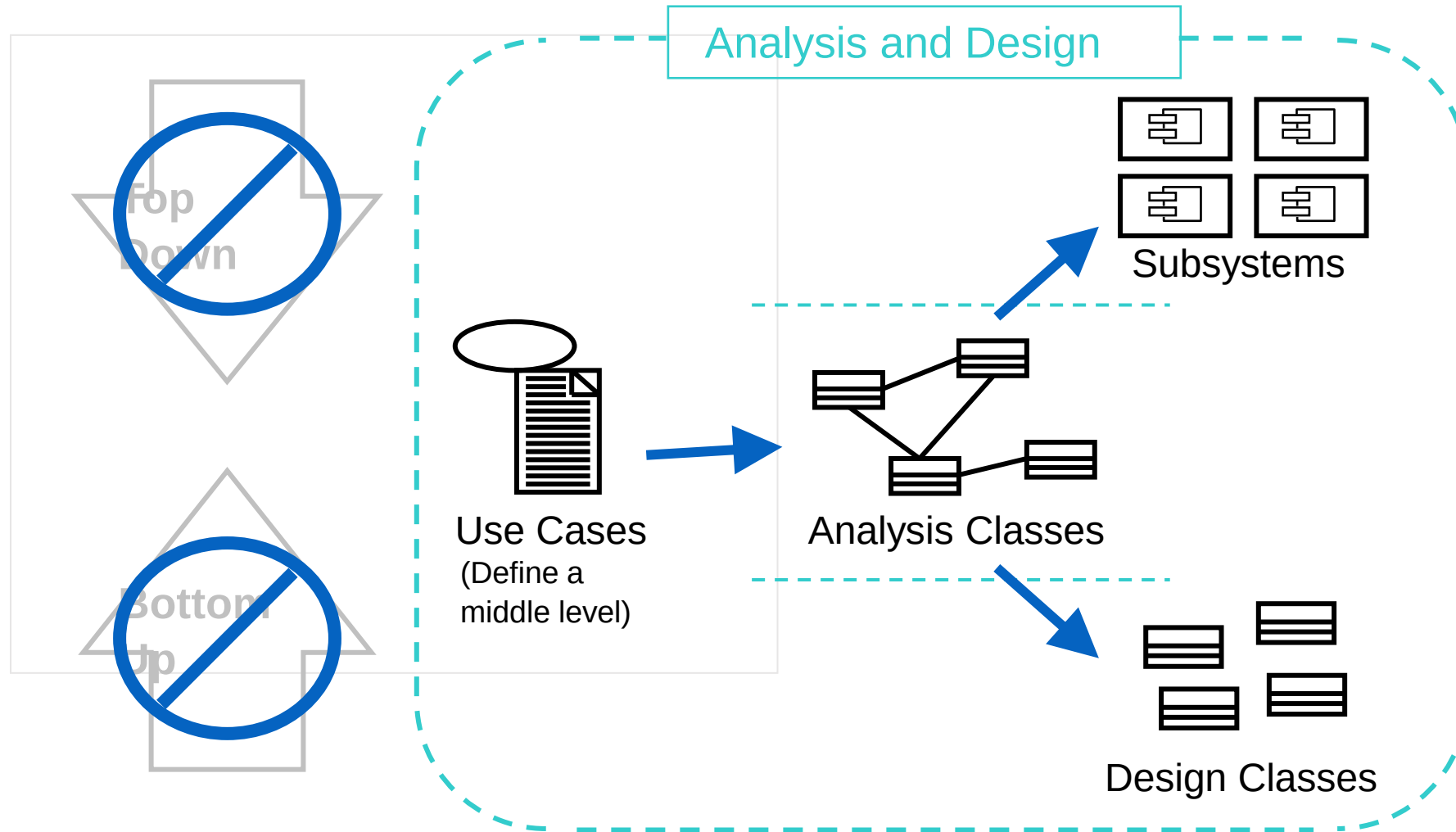


Analysis versus Design



Analysis	Design
■ Focus on understanding the problem ■ Idealized design ■ Behavior ■ System structure ■ Functional requirements ■ A small model	■ Focus on understanding the solution ■ Operations and attributes ■ Performance ■ Close to real code ■ Object lifecycles ■ Nonfunctional requirements ■ A large model

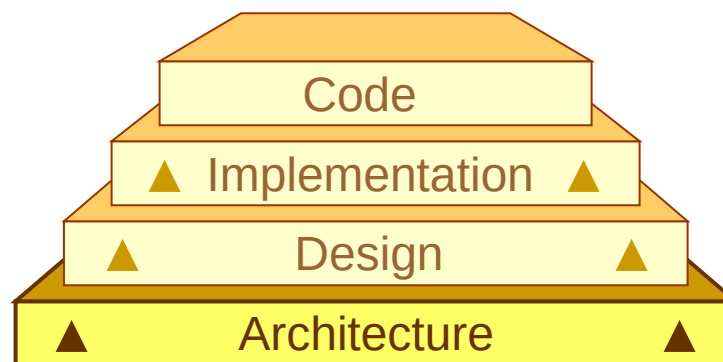
Analysis and Design Are Not Top-Down or Bottom-Up



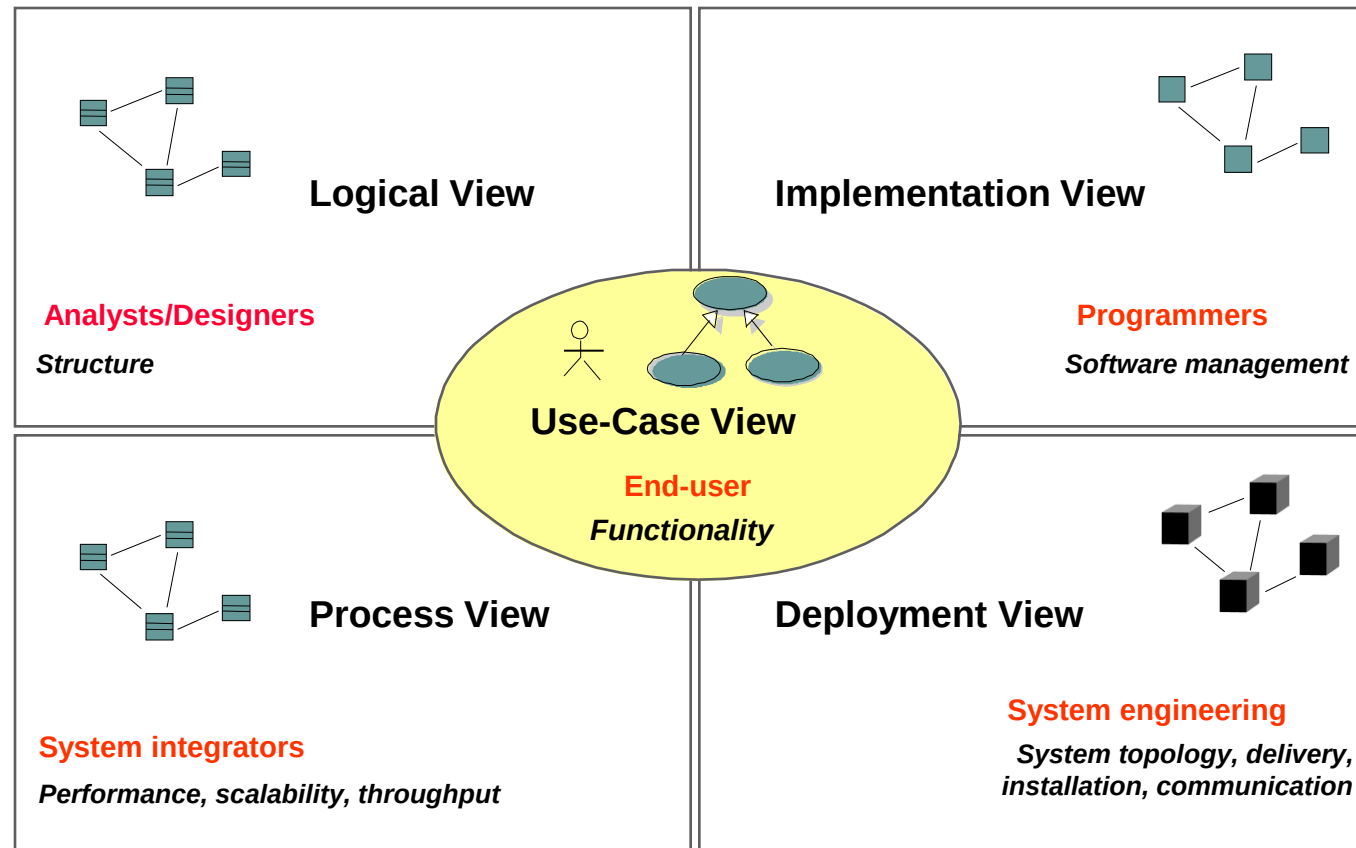
Software architecture



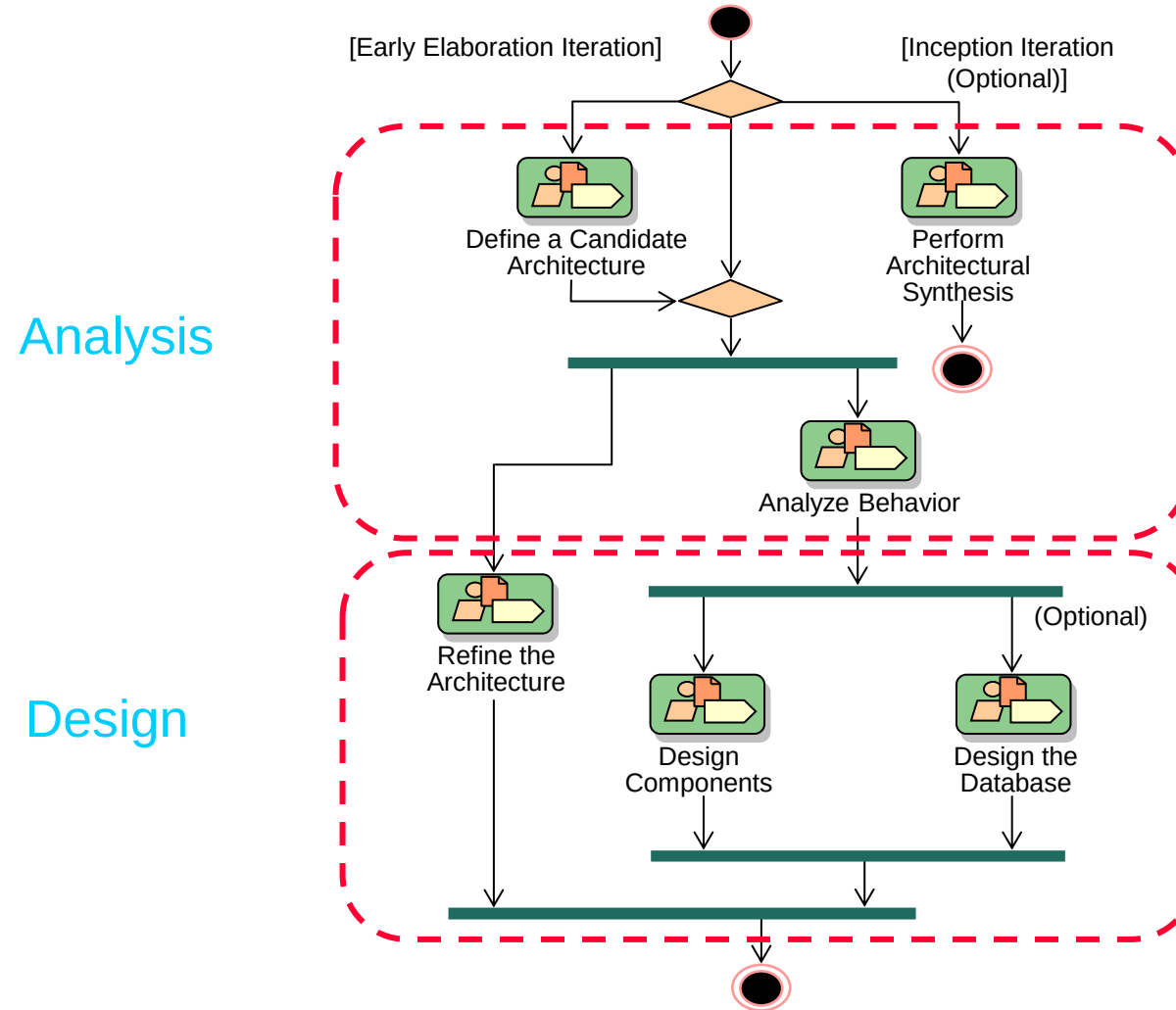
- ❖ Software architecture encompasses a set of significant decisions about the organization of a software system
- ❖ Architecture involves a set of strategic design decisions, rules or patterns that constrain design and construction.



Software architecture: The “4+1 View” model



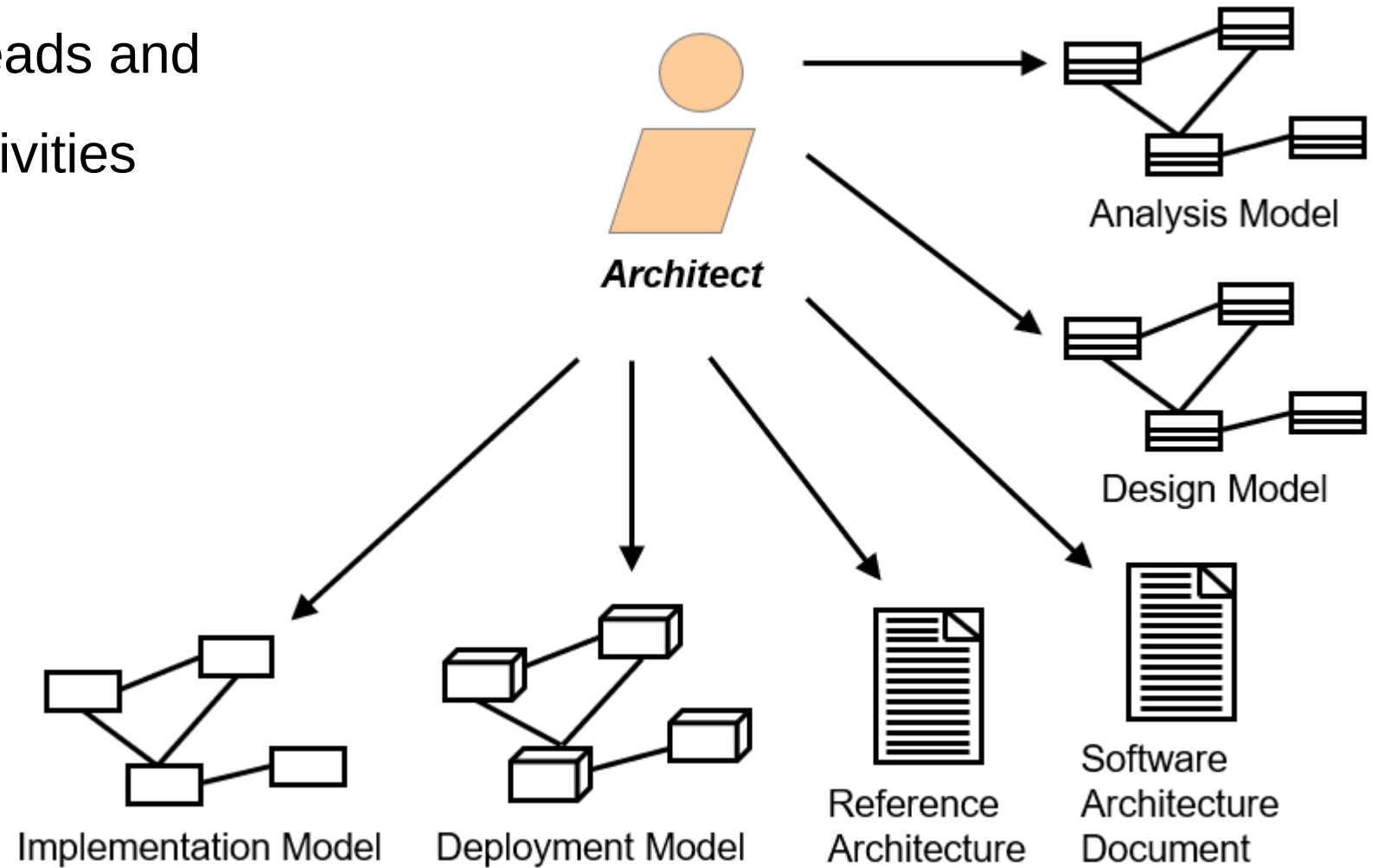
Analysis and Design Workflow



Software Architect's Responsibilities



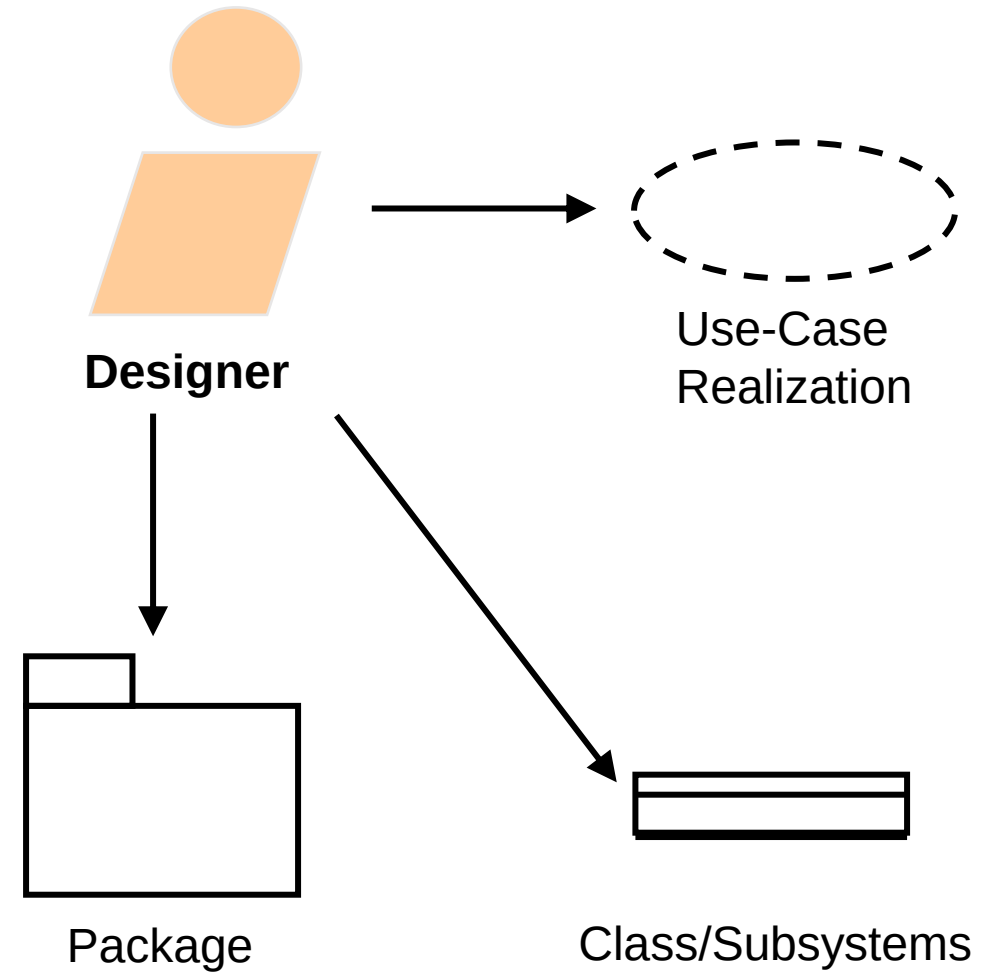
- ❖ The Software Architect leads and coordinates technical activities and artifacts



Designer's Responsibilities



- ❖ The designer must know use-case modeling techniques, system requirements, and software design techniques.

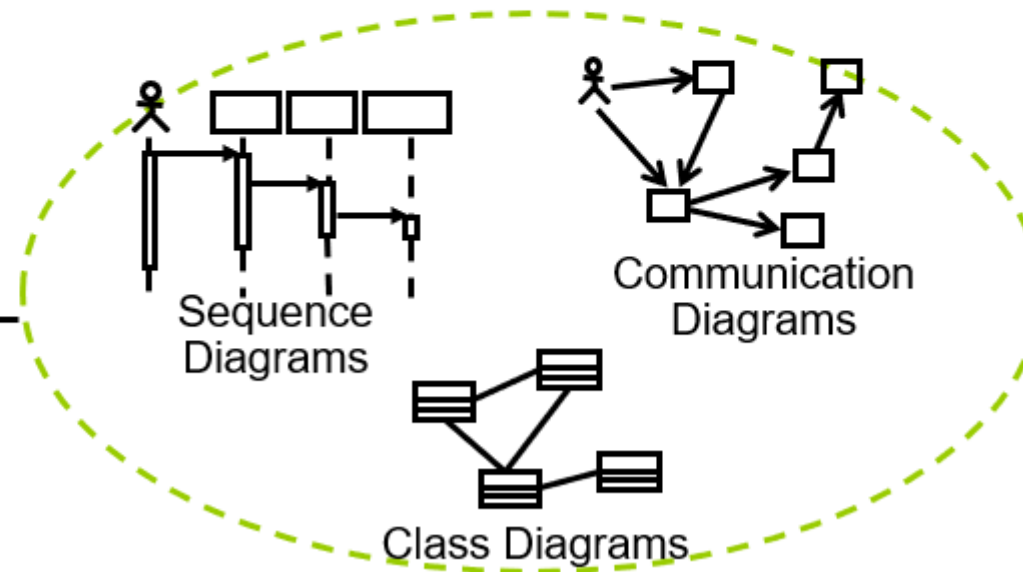


Use Case Realization

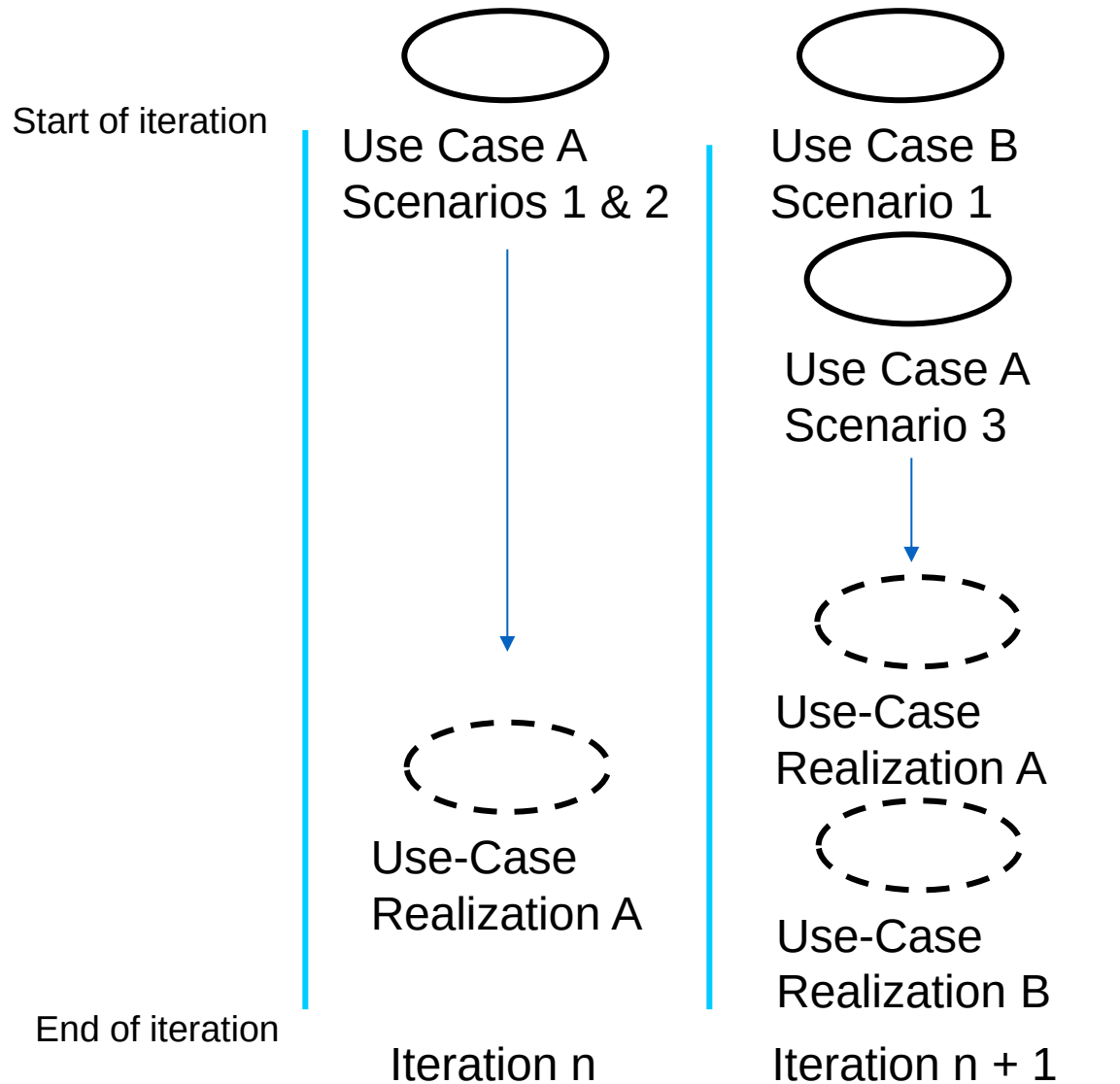


Use-Case Model

Design Model



Analysis and Design in an Iterative Process

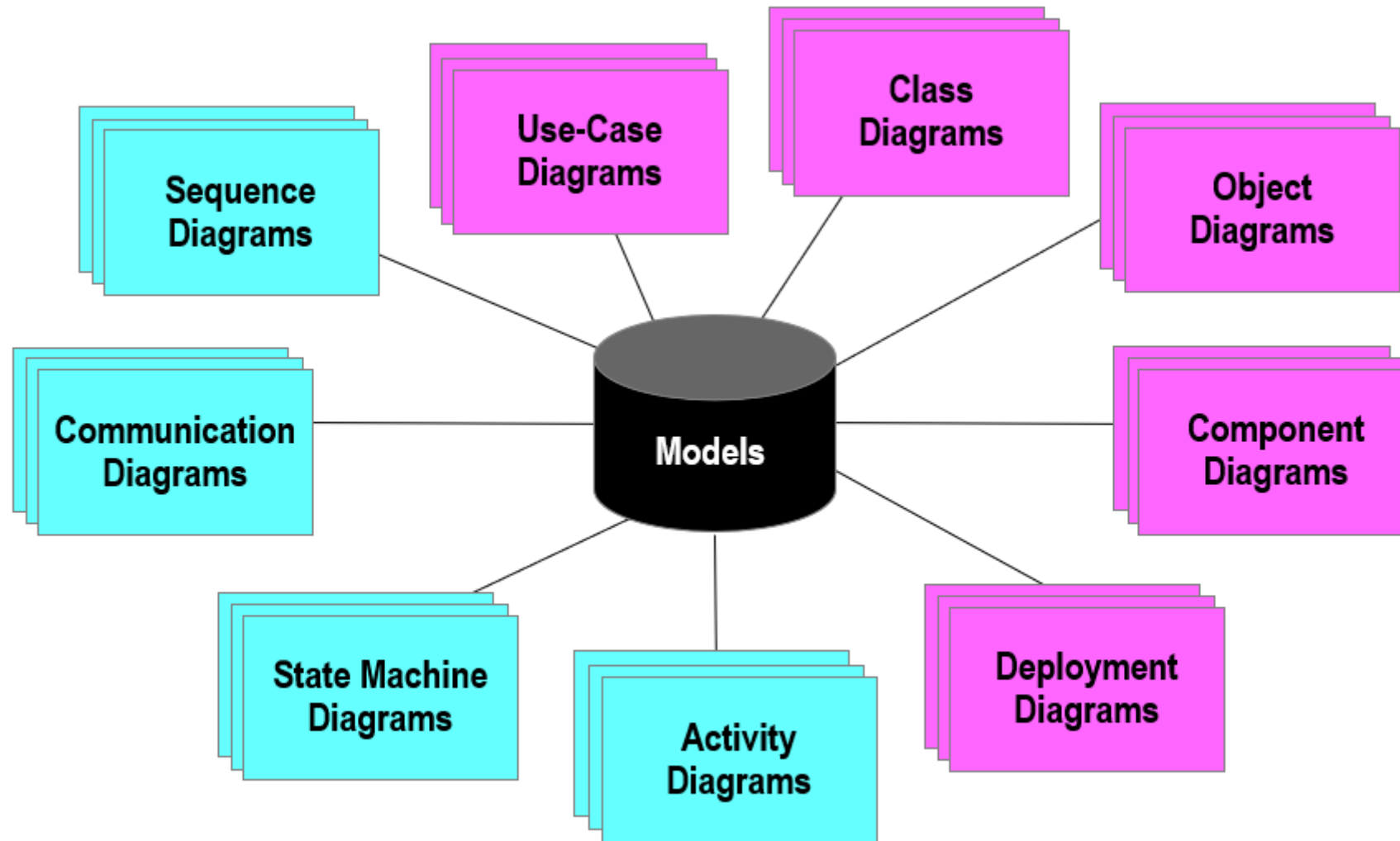


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Visual Modeling with the UML





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Q&A